5 Results

5.1 Manufactured Composite Aerofoil

The manufactured part is a composite aerofoil section made of two cured hybrid carbon/glass skins, forming the upper and lower aerodynamic surfaces, bonded to a polyurethane (PU) foam core (Figure 3). No dimensional or mass measurements were taken during the laboratory, so the part is described qualitatively and its composition is estimated from nominal material data (Section 5.3). Its geometry is set by the tooling rather than by the operator: the aluminium moulds define the outer aerofoil surface, the foam core defines the internal profile, and the skin thickness is set by the number of plies and the consolidation achieved under vacuum. Both skins were cured in a single oven load together with the other bagged moulds from the session (Figure 4a).

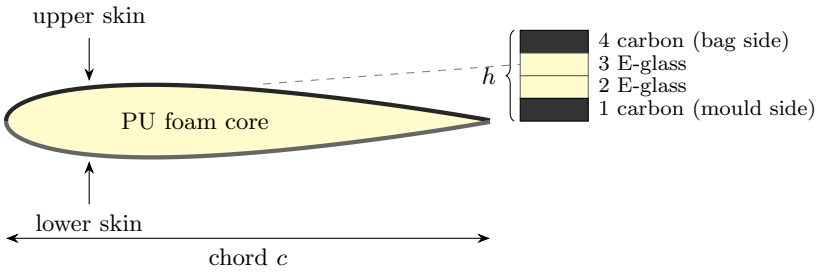


Figure 3: Schematic section of the aerofoil (not to scale): two skins bonded to the PU foam core, each with the $[\mathrm{C}/\mathrm{G}]_\mathrm{s}$ ply sequence shown on the right.

5.2 Laminate Architecture

Each skin is a four-ply, symmetric, inter-ply hybrid laminate (Figure 3, Table 3). Ply 1, laid against the mould and forming the aerodynamic surface, and ply 4, on the bag side, are woven carbon/epoxy prepreg; plies 2 and 3 at the mid-plane are woven E-glass/epoxy prepreg. With the warp of every fabric aligned with the span, the stacking sequence is $[0_\mathrm{C}/0_\mathrm{G}]_\mathrm{s}$, where each 0 denotes a 0/90° woven ply **[TBC: confirm ply angles and fabric styles]**. The two skins together contain eight plies, four carbon and four glass **[TBC: confirm four plies per skin]**. The lay-up is symmetric and balanced, so neither bending–extension nor extension–shear coupling is expected, and the skins should not warp on cooling from the cure temperature.

Table 3: Laminate architecture of each skin (A_w = fabric areal mass, ρ_f = fibre density, W_r = prepreg resin content by mass). *Nominal values typical of woven epoxy prepregs **[TBC: replace with data-sheet values if known]**.

Ply	Position	Prepreg	Angle	A_w ($\mathrm{g\,m^{-2}}$)	ρ_f ($\mathrm{g\,cm^{-3}}$)	W_r (wt%)
1	Mould side	Carbon/epoxy, woven	0/90	200*	1.76	40*
2	Mid-plane	E-glass/epoxy, woven	0/90	200*	2.54	35*
3	Mid-plane	E-glass/epoxy, woven	0/90	200*	2.54	35*
4	Bag side	Carbon/epoxy, woven	0/90	200*	1.76	40*

5.3 Material Composition

The fibre densities are $1.76\,\mathrm{g\,cm^{-3}}$ for the T300-type carbon [1] and $2.54\,\mathrm{g\,cm^{-3}}$ for E-glass [2], and the cured epoxy matrix is about $1.2\,\mathrm{g\,cm^{-3}}$ [2]. With equal nominal areal masses of $200\,\mathrm{g\,m^{-2}}$, carbon and glass each make up half of the plies and half of the fibre mass, but because carbon is about 30 % less dense it makes up about 59 % of the fibre volume (glass 41 %).

Since the cured skins were not measured, their composition was estimated theoretically. For a void-free laminate with no resin bleed, the cured thickness of each ply follows from its areal mass and

prepreg resin content:

$$t_{\mathrm{ply},i} = \frac{A_{w,i}}{\rho_{f,i}} + \frac{A_{w,i}}{\rho_m}\,\frac{W_{r,i}}{1 - W_{r,i}}, \tag{1}$$

where the first term is the thickness of solid fibre and the second that of resin. The fibre volume fraction of the laminate is then

$$V_f = \frac{1}{h}\sum_i \frac{n_i A_{w,i}}{\rho_{f,i}}, \tag{2}$$

where n_i is the number of plies of fabric i and h the laminate thickness [3]. The results (Table 4) give a skin thickness of 0.786 mm and $V_f \approx 49\,\%$, made up of 28.9 % carbon and 20.0 % glass fibre, with 51.1 % matrix by volume. These are ideal values: in the real skins, resin bleed into the peel ply and breather would raise V_f, while voids and incomplete consolidation under vacuum alone would lower it. Because the summed fibre thickness is 0.385 mm, Eq. (2) reduces to $V_f = 0.385/h$ (h in mm), so a single thickness measurement would be enough to check the estimate.

Table 4: Theoretical (void-free, no resin bleed) composition of each skin, from Eqs. (1) and (2) with the nominal values in Table 3.

	Thickness (mm)	V_f (%)	**Fibre mass fraction** (%)	**Density** ($\mathrm{g\,cm^{-3}}$)	**Areal mass** ($\mathrm{g\,m^{-2}}$)
Carbon/epoxy ply	0.225	50.6	60.0	1.48	333
E-glass/epoxy ply	0.168	46.7	65.0	1.83	308
Skin $[\mathrm{C/G}]_\mathrm{s}$	0.786	48.9	62.4	1.63	1282

5.4 Surface Quality and Manufacturing Defects

No inspection of the cured skins was recorded **[TBC: add any visual observations or a photograph of the cured skins]**, but their surface quality follows from the tooling and consumables used. The mould-side faces, which form the aerodynamic surface, replicate the finish of the polished aluminium moulds and show the carbon weave [4, 5], whereas the bag-side faces carry the uniform matte imprint of the peel ply, which leaves a clean, textured surface suited to adhesive bonding to the core [6]. Table 5 lists the defects most likely for this process, where they would appear and why; they are discussed further in the Discussion.

Table 5: Manufacturing defects most likely in the vacuum-bagged skins.

Defect	**Most likely location**	**Cause in this process**
Pinholes / surface porosity	Mould (tool) side	Air trapped at the tool surface and between plies; vacuum-only consolidation [7–9]
Dry or resin-starved areas	Bag side, under the breather	Excess resin absorbed by the peel ply and breather [10, 11]
Ply bridging / resin-rich or porous corners	Tight concave radii of the mould, e.g. leading edge	Plies span the radius instead of seating on it, so local compaction pressure is low [12]
Wrinkles (out-of-plane waviness)	Curved regions	Ply path-length differences and non-uniform pressure on curved surfaces [13, 14]
Fibre misalignment	Whole skin	Manual cutting and placement of plies [14, 15]
Frayed edges / edge delamination	Cut edges	Trimming of the cured laminate [16]
Warping	Whole skin (slight)	Not expected from the symmetric lay-up [2], but tool–part interaction with the aluminium mould can still distort thin skins [17, 18]

5.5 Curing Temperature Profile

Figure 4 shows the curing set-up. The bagged moulds were cured on the floor of a fan-assisted (forced-convection) oven together with the other groups' moulds. The bags' vacuum ports were connected by

flexible tubing and quick-release couplings to a vacuum line that passes through the oven wall to an Edwards oil-sealed rotary vane vacuum pump fitted with an exhaust oil-mist filter. The pump kept the bags evacuated throughout the cycle, so the laminates were consolidated by atmospheric pressure (at most ≈101 kPa) without an autoclave.



(a) (b)

Figure 4: Curing set-up: (a) bagged moulds in the fan-assisted oven, with the vacuum tubing connected to a bag's vacuum port through a quick-release coupling; (b) the rotary vane vacuum pump, connected to the oven's vacuum feed-through.

The oven ran the four-segment programme 'MECH2460' (Figure 5, Table 6). It ramped from room temperature to 60 °C over 35 min (≈1.1 °C/min, taking room temperature as ≈20 °C), held at 60 °C for 60 min, ramped to 120 °C over 30 min (2.0 °C/min) and held at 120 °C for 2 h, a programmed cycle of 4 h 5 min. The event outputs (TS1–TS3) and the wait function were set to zero, so each segment ran for its set time whatever the actual chamber temperature. No cooling segment was programmed: the parts cooled with the oven, still under vacuum, before the bags were opened (Section 4, step 11).

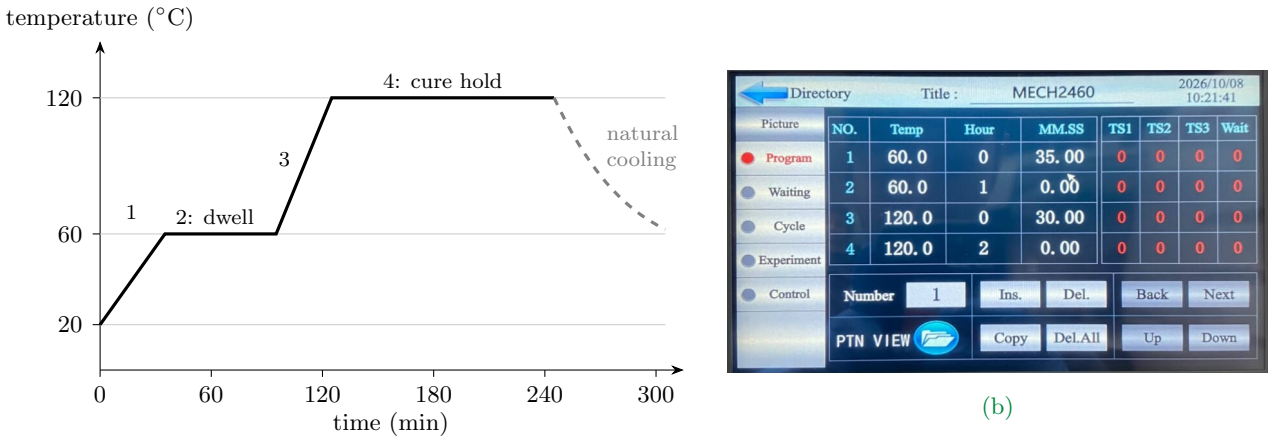
This two-stage cycle is typical of epoxy prepregs cured under vacuum alone. The 60 °C dwell lowers the resin viscosity and holds the resin fluid, so the plies can wet out and consolidate and air trapped between them has time to escape through the breather before the resin gels. The faster second ramp then brings the laminate to 120 °C, where the 2 h hold completes cross-linking and sets the glass-transition temperature [8, 11]. The moderate ramp rates limit thermal gradients across the aluminium moulds, whose thermal mass makes the laminate temperature lag the oven air temperature. Because the part temperature was not monitored with a thermocouple, the time the laminate actually spent at 120 °C was somewhat shorter than the programmed 2 h.

Table 6: Oven programme 'MECH2460' used to cure the skins.

Segment	**Target** (°C)	**Duration**	**Rate** (°C/min)	**Elapsed**	**Purpose**
1	60	35 min	≈1.1	0:35	Heat-up
2	60	60 min	0 (dwell)	1:35	Resin flow, consolidation, air removal
3	120	30 min	2.0	2:05	Heat-up to cure temperature
4	120	2 h	0 (hold)	4:05	Cure
–	–	–	–	–	Natural cooling under vacuum

5.6 Final Assembly and Bonding

After cooling, the bags were cut open, the breather and peel ply were removed and the skins were released from the moulds (Section 4). Stripping the peel ply exposed the textured bag-side surfaces, and the two skins were then placed over the pre-shaped PU foam core and bonded with **[TBC: adhesive**



(a) (b)

Figure 5: Curing temperature profile: (a) programmed oven set-point (solid; room temperature taken as 20 °C) and the unprogrammed natural cool-down (dashed, indicative only); (b) the programme on the oven controller.

used] [TBC: confirm whether the bonding was done in the session or afterwards]. The adhesive bond line adds to the section thickness, and any mismatch between the skins and the core appears as gaps or steps at the leading and trailing edges, so the assembled aerofoil is expected to be slightly thicker than the nominal section formed by the core and two 0.786 mm skins.

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